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Project Title:	Neural Processing of Ambiguous Gravito-inertial Cues: A Cross Species Comparison		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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	To learn how the nervous system processes ambiguous gravitational and inertial cues such that veridical representations of acceleration and orientation can be calculated by the brain and the appropriate motor and perceptual responses generated. To compare perceptual and eye movement responses in rhesus monkeys and human subjects during motion paradigms that presesent ambiguous information from the vestibular labyrinth regarding the subjects spatial orientation with respect to gravity and linear acceleration Sensory information from the otolith organs is inherently ambiguous, yet both humans and non-human primates are able to make fairly accurate perceptual estimates of head orientation and motion as well as reflexive eye movements when presented with these ambiguous cues. While non-human primates are widely used as model systems to study vestibular physiology that is presumably relevant to human function, it is clear from prior work that vestibular-mediated responses differ substantially between monkeys and humans. The work we		

Task Description:	have performed has helped to identify the mechanisms used by the brain to perform this tilt/translation discrimination in both species, and has therefore helped to elucidate the basis of the differences between human and monkey responses. The overall findings are that, in monkeys, both eye movement and perceptual responses appear to be derived from a process of sensory integration, that combines gravito-inertial cues sensed by the otolith organs with rotational cues sensed by the semicircular canals and visual system. In contrast, in humans perceptual responses appear to be produced by similar mechanisms of sensory integration, but eye movement responses appear to result from a simpler mode of processing that segregates tilt from translation with a frequency criteria. In human eye movements, a simple filtering process appears to be used such that torsional (appropriate for tilt) response occurs during low frequency otolith stimulation, while horizontal (appropriate for translation) eye movements occur with higher frequency otolith stimuli.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work helps us understand why spatial disoreintation can occur on earth in normal human subjects when they are presented with ambiguous vestibular cues, and why patients with vestibular disorders (which generate incongruent vestibular signals) produce spatial disorientation. It helds to understand why subjects who have experienced environments with altered gravitational force (e.g. in space) suffer from ataxia and spatial disorientation when returning to the 1-G environment of earth.
Task Progress:	During the three year period of this grant, we accomplished a number of goals. We developed a method to measure roll tilt psychophysics in rhesus monkeys; used this method to measure perceptual responses in monkeys during a number of motion paradigms that present the brain with ambiguous vestibular cues (rotational information supplied by the canals and shifts in gravito-inertial force [GIF] orientation supplied by the otoliths); and compared the perceptual responses with three dimensional eye movements measured during the same motion paradigms. In human subjects, we also measured tilt perception and eye movements during translational and tilt motion in order to compare and contrast rhesus and human psychophysical and oculomotor responses. The tilt psychophysical method developed for non-human primates was based on the subjective visual vertical (SVV) test widely used in humans for many years. Two monkeys were trained to turn a small steering wheel attached to the primate chair to rotate a light bar so that they percieved its orientation to be parallel to the earth vertical. The task was taught to the animals with extensive training procedures described elsewhere. After about 6 months of training, the monkeys were able to accurately align the light bar with the gravitational vector in the absence of any visual cues. Once the monkeys were adequately trained, we tested their perception of roll tilt during a number of motion protocols. Three dimensional eye movements were also measured (in the dark) during the same protocols. Motions included: static roll tilt, dyanamic roll tilt; static tilt with roll OKN stimulation; inter-aural (IA) translation, and centrifugation with the rotational arm aligned with the IA axis. The results are summarized below: Roll tilt: During static roll tilt, the monkeys accurately estimated head orientation in the roll plane. There was evidence of mild adaptation during sustained tilts (e.g. response decay followed by overshoot when returning to the upright position). No consistent A or E effects were noted. During dynamic roll tilt, the animals' responses remained closely aligned with the gravitational vector. When roll OKN stimulation was added during static roll tilt, the perceived head orientation was deviated in the opposite direction of the OKN motion in a manner that depended on head orientation. IA translation: Testing was limited to three low frequencies(0.01, 0.05, and 0.1 Hz) given the nature of the perceptual task. At these frequencies, IA translation was primarily perceived as roll tilt (tilt gains, defined as maximum perceived tilt/maximum tilt of the GIF vector) were approximately 0.8). Centrifugation: In the standard fixed-radius (the animal is placed eccentrically and then accelerated in yaw rotation to a constant angular velocity, we found that perceived tilt built up slowly, but eventually approximated the tilt of the GIF vector. Conversely, during variable-radius centrifugation (the animal is initially accelerated to a constant yaw angular velocity and then translated eccentrically), perceived tilt changed rapidly and nearly remained aligned with GIF tilt, although it did lag slightly. These results, which represent the first tilt psychophysical measurements made in non-human primates, are consistent with the sensory integration hypothesis of tilt-translation discrimination. Briefly, percieved tilt most closely aligned with the GIF when otolith and canal rotational cues were congruent (e.g. during true roll tilt), percieved tilt lagged slightly behind tilt of the GIF when canal cues were absent but the GIF shifted in orientation (e.g. IA translation and variable-radius centrifugation), and lagged most dramatically when the canal rotational cue (in yaw) was orthogonal to the roll shift in GIF orientation (fixed-axis centrifugation). Eye movements measured in the monkeys showed similar evidence of sensory integration as the underlying mechanism used to discriminate tilt from translation. Torsional eye movements (indicative of roll tlt) were most prominant during true roll tilts but were also observed during low frequency IA translation. The eye rotational axis aligned with gravity slowly during fixed-radius centrifugation, similar to prior reports in monkeys. Overall, the psychophysical and eye movement results in rhesus monkeys suggest that a mechanism of sensory integration, which combines rotational cues derived from the canals with gravitational and inertial cues sensed by the otoliths, is used to generate perceptions of roll tilt and reflexive vestibular eye movements during head motion. In human subjects, during tilt, translation and combined tilt/translation, we similarly found that perceived roll tilt depended on a synthesis of canal and otolith cues. Conversely, we found that the eye movements produced by human subjects were dependent on the frequency of the shift in GIF orientation but not on an integration of canal and otolith information. For example, similar eye movements were generated during tilts and translations when the frequency of the motion was matched, and during combined tilt/translation, the net response simply reflected the net shift in GIF orientation. In sum, these experiments demonstrate that rhesus monkeys use an integrative mechanism that combines canal and otolith cues to generate both percpeptual and eye movement responses during head motion, while humans use a similar mechanism to generate perceptual responses, but a simpler filtering method to generate eye movement responses. The basis for the difference in eye movements between rhesus and humans is uncertain, but may related to central velocity storage mechanisms. Specifically, it has been suggested that, at least for eye movements, velocity storage is the substrate for integrating canal and otolith information. Since velocity storage is more prominent in monkeys (compared to humans), the difference in eye movement responses may reflect the stronger velocity storage in monkeys (leading to sensory integration as the operative mechanism to generate eye movements) and weaker velocity storage in humans

	(leading to a simpler, frequency-specific mechanism to generate eye movements). This hypothesis could be tested with further research that alters velocity storage in monkeys, either chemically (e.g. with GABA agonists such as baclofen) or via structural lesions (e.g. midline brainstem lesions that disable velocity storage).
Bibliography Type:	Description: (Last Updated: 05/23/2006)
Articles in Peer-reviewed Journals	Lewis RF, Haburcakova C, Merfeld MD. "Tilt psychophysics measured in non-human primates." Ann N Y Acad Sci. 2005 Apr;1039:294-305. PMID: 15826983 , Apr-2005
Articles in Peer-reviewed Journals	Merfeld DM, Park S, Gianna-Poulin C, Black FO, Wood S. "Vestibular perception and action employ qualitatively different mechanisms. I. Frequency response of VOR and perceptual responses during Translation and Tilt." J Neurophysiol. 2005 Jul;94(1):186-98. PMID: 15728767 , Jul-2005